

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

Q.1 Create two 200MB partitions using fdisk or parted, then use the lsblk command to verify their creation.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 17:31
root@localhost:~

Partition number (1-4, default 1):
First sector (2048-4194303, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303): +2G
Value out of range.
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-4194303, default 4194303):

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): p
Disk /dev/nvme0n2: 2 GiB, 2147483648 bytes, 4194304 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x317a0d10

Device          Boot Start      End Sectors Size Id Type
/dev/nvme0n2p1   2048 4194303 4192256  2G 83 Linux

Command (m for help): q

[root@localhost ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0         11:0    1 10.8G  0 rom  /run/media/root/CentOS-Stream-9-BaseOS-x86_64
nvme0n1     259:0    0   40G  0 disk 
├─nvme0n1p1 259:1    0    1G  0 part /boot
├─nvme0n1p2 259:2    0   39G  0 part 
├─cs-root    253:0    0  35.1G  0 lvm  /
├─cs-swap    253:1    0   3.9G  0 lvm  [SWAP]
nvme0n2     259:3    0    2G  0 disk 
├─nvme0n2p1 259:4    0  200M  0 part 
└─nvme0n2p2 259:5    0  200M  0 part
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 17:45
root@localhost:~

Created a new partition 2 of type 'Linux filesystem' and of size 200 MiB.

Command (m for help): p
Disk /dev/nvme0n2: 2 GiB, 2147483648 bytes, 4194304 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 7004FE16-7EC2-FE49-B392-A6EA098D56C1

Device          Start      End Sectors Size Type
/dev/nvme0n2p1   2048 411647  409600 200M Linux filesystem
/dev/nvme0n2p2 411648 821247  409600 200M Linux filesystem

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

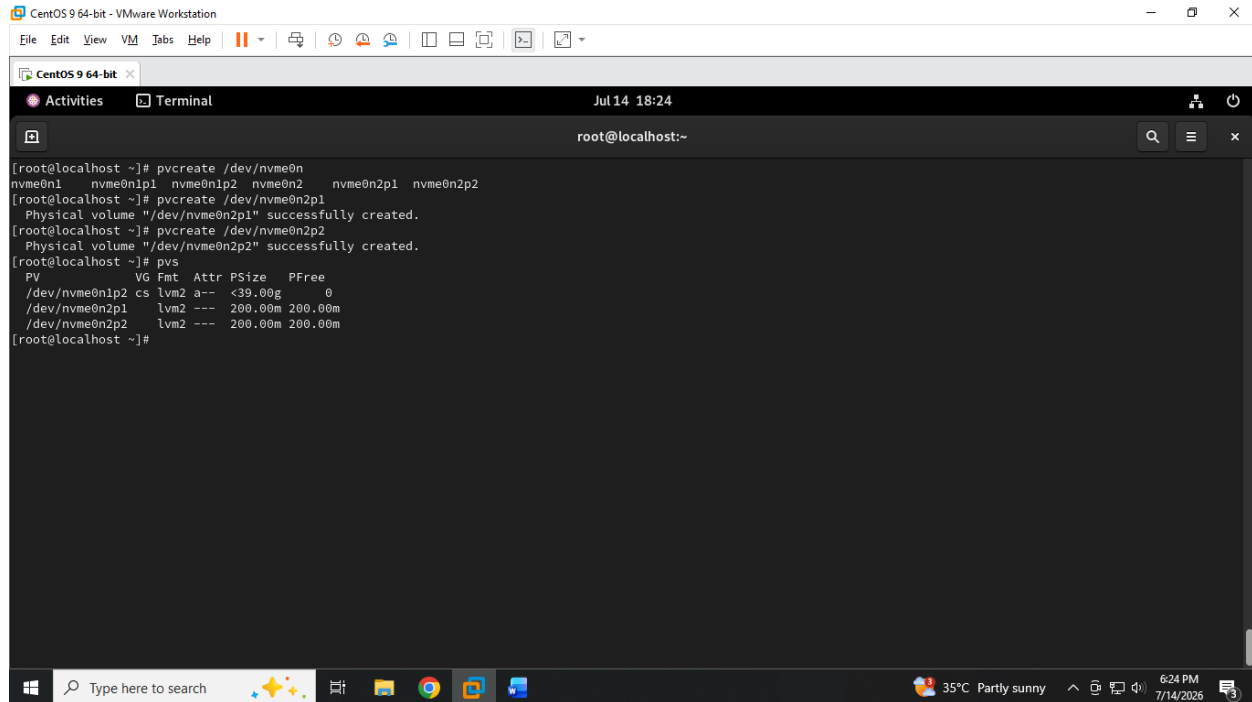
[root@localhost ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0         11:0    1 10.8G  0 rom  /run/media/root/CentOS-Stream-9-BaseOS-x86_64
nvme0n1     259:0    0   40G  0 disk 
├─nvme0n1p1 259:1    0    1G  0 part /boot
├─nvme0n1p2 259:2    0   39G  0 part 
├─cs-root    253:0    0  35.1G  0 lvm  /
├─cs-swap    253:1    0   3.9G  0 lvm  [SWAP]
nvme0n2     259:3    0    2G  0 disk 
├─nvme0n2p1 259:4    0  200M  0 part 
└─nvme0n2p2 259:5    0  200M  0 part
```

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

Q.2 Initialize both partitions as physical volumes using the pvcreate command and confirm their status with pvs.

ANS :



The screenshot shows a terminal window titled 'CentOS 9 64-bit' within a VMware Workstation environment. The terminal displays the following commands and output:

```
[root@localhost ~]# pvcreate /dev/nvme0n1
nvme0n1  nvme0n1p1 nvme0n1p2 nvme0n2  nvme0n2p1 nvme0n2p2
[root@localhost ~]# pvcreate /dev/nvme0n2p1
Physical volume "/dev/nvme0n2p1" successfully created.
[root@localhost ~]# pvcreate /dev/nvme0n2p2
Physical volume "/dev/nvme0n2p2" successfully created.
[root@localhost ~]# pvs
```

PV	VG	Fmt	Attr	PSize	PFree
/dev/nvme0n1p2	cs	lvm2	a--	<39.00g	0
/dev/nvme0n2p1		lvm2	---	200.00m	200.00m
/dev/nvme0n2p2		lvm2	---	200.00m	200.00m

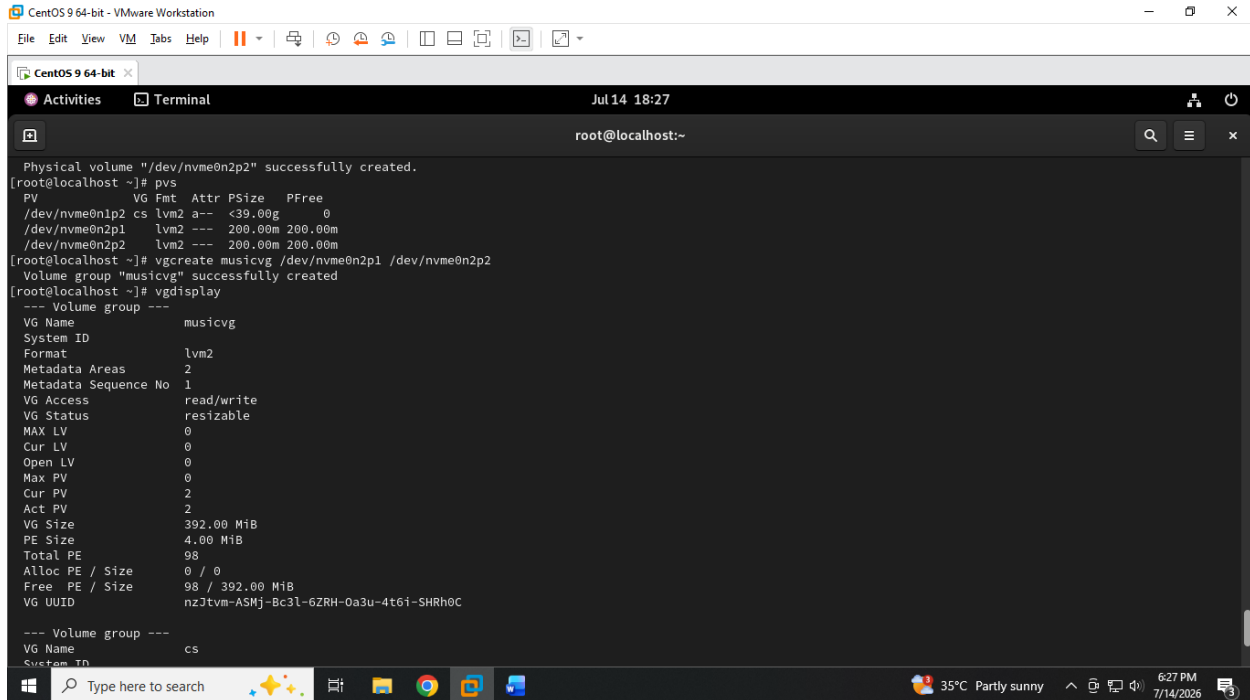
The terminal window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status information (35°C, Partly sunny, 6:24 PM, 7/14/2026).

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

Q.3 Create a new volume group named musicvg using the two physical volumes, then display its details with vgdisplay.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:27
root@localhost:~

Physical volume "/dev/nvme0n2p2" successfully created.
[root@localhost ~]# pvs
PV          VG Fmt Attr PSize  PFree
/dev/nvme0n1p2 cs  lvm2 a--  <39.00g    0
/dev/nvme0n2p1 lvm2 ---  200.00m  200.00m
/dev/nvme0n2p2 lvm2 ---  200.00m  200.00m
[root@localhost ~]# vgcreate musicvg /dev/nvme0n2p1 /dev/nvme0n2p2
Volume group "musicvg" successfully created
[root@localhost ~]# vgdisplay
--- Volume group ---
VG Name                musicvg
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   1
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                 0
Cur PV                 2
Act PV                 2
VG Size                392.00 MiB
PE Size                4.00 MiB
Total PE                98
Alloc PE / Size        0 / 0
Free PE / Size         98 / 392.00 MiB
VG UUID                nzJtvm-ASMj-Bc3L-6ZRH-0a3u-4t6I-SHRh0C

--- Volume group ---
VG Name                cs
System ID
```

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:24
root@localhost:~

[root@localhost ~]# pvcreate /dev/nvme0n
nvme0n1 nvme0n1p1 nvme0n1p2 nvme0n2 nvme0n2p1 nvme0n2p2
[root@localhost ~]# pvcreate /dev/nvme0n2p1
Physical volume "/dev/nvme0n2p1" successfully created.
[root@localhost ~]# pvcreate /dev/nvme0n2p2
Physical volume "/dev/nvme0n2p2" successfully created.
[root@localhost ~]# pvs
PV VG Fmt Attr PSize PFree
/dev/nvme0n1p2 cs lvm2 a-- <39.00g 0
/dev/nvme0n2p1 lvm2 --- 200.00m 200.00m
/dev/nvme0n2p2 lvm2 --- 200.00m 200.00m
[root@localhost ~]#
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:27
root@localhost:~

Max PV 0
Cur PV 2
Act PV 2
VG Size 392.00 MiB
PE Size 4.00 MiB
Total PE 98
Alloc PE / Size 0 / 0
Free PE / Size 98 / 392.00 MiB
VG UUID nzJtvm-ASMj-Bc3L-6ZRH-Oa3u-4t61-SHRh0C

--- Volume group ---
VG Name cs
System ID
Format lvm2
Metadata Areas 1
Metadata Sequence No 3
VG Access read/write
VG Status resizable
MAX LV 0
Cur LV 2
Open LV 2
Max PV 0
Cur PV 1
Act PV 1
VG Size <39.00 GiB
PE Size 4.00 MiB
Total PE 9983
Alloc PE / Size 9983 / <39.00 GiB
Free PE / Size 0 / 0
VG UUID JnorUh-g0dT-7Ado-Kd0t-R2gc-usHa-MM35o9

[root@localhost ~]#
```

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

Q.4 Create a logical volume called playlistlv of size 300MB inside musicvg, format it with ext4, and mount it to /mnt/playlist.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:35
root@localhost:~

[root@localhost ~]# lvcreate -L 300M -n playlistlv musicvg
Logical volume "playlistlv" created.
[root@localhost ~]# lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
root cs -wi-ao---- <35.08g
swap cs -wi-ao---- <3.92g
playlistlv musicvg -wi-a----- 300.00m
[root@localhost ~]# e
e2freefrag ebttables edquota encguess era_check eutp exit
e2fsck ebttables-nft egk-tool enchant-2 era_dump eval exiv2
e2image ebttables-nft-restore egrep enchant-lsmo-2 era_invalidate evince expand
e2label ebttables-nft-save eidenv encrypt era_restore evince-preview export
e2mmpstatus ebttables-restore eject env esac evince-thumbnail expr
e2undo ebttables-save elfedit envsubst escputil evmctl
e4crypt ebttables-translate elif eog espeak-ng ex
e4defrag echo else eog ether-wake exec
eapol_test ed enable eqn ethtool exempi

[root@localhost ~]# mkfs.ext4 /dev/musicvg/playlistlv
mkfs2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: 320bd27d-0dc4-4633-ae2d-b0a15083545b
Superblock backups stored on blocks:
8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]#
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:36
root@localhost:~

e2image ebttables-nft-restore egrep enchant-lsmo-2 era_invalidate evince expand
e2label ebttables-nft-save eidenv encrypt era_restore evince-preview export
e2mmpstatus ebttables-restore eject env esac evince-thumbnail expr
e2undo ebttables-save elfedit envsubst escputil evmctl
e4crypt ebttables-translate elif eog espeak-ng ex
e4defrag echo else eog ether-wake exec
eapol_test ed enable eqn ethtool exempi

[root@localhost ~]# mkfs.ext4 /dev/musicvg/playlistlv
mkfs2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: 320bd27d-0dc4-4633-ae2d-b0a15083545b
Superblock backups stored on blocks:
8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

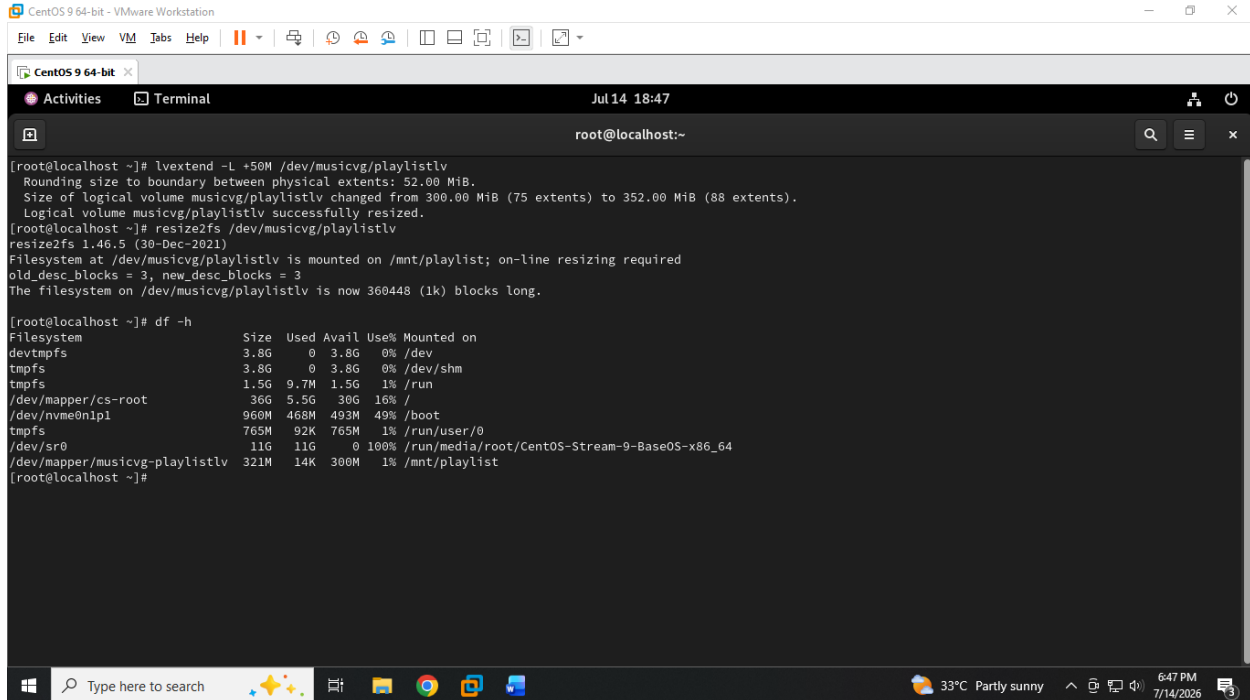
[root@localhost ~]# mkdir /mnt/playlist
[root@localhost ~]# mount /dev/musicvg/playlistlv /mnt/playlist/
[root@localhost ~]# df -h
Filesystem Size Used Avail Use% Mounted on
devtmpfs 3.8G 0 3.8G 0% /dev
tmpfs 3.8G 0 3.8G 0% /dev/shm
tmpfs 1.5G 9.7M 1.5G 1% /run
/dev/mapper/cs-root 36G 5.5G 30G 16% /
/dev/nvme0n1p1 960M 468M 493M 49% /boot
tmpfs 765M 92K 765M 1% /run/user/0
/dev/sr0 11G 11G 0 100% /run/media/root/CentOS-Stream-9-Base05-x86_64
/dev/mapper/musicvg-playlistlv 272M 14K 253M 1% /mnt/playlist
[root@localhost ~]#
```

MODULE-5 : LOGICAL VOLUME MANAGEMENT (LVM)

GUIDED BY : VIPUL MISTRI

Q.5 Resize playlistlv by adding 50MB more space using lvextend, then expand the filesystem and show the updated size with df -h. Hint: Use resize2fs after lvextend to update the filesystem size.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 14 18:47
root@localhost:~

[root@localhost ~]# lvextend -L +50M /dev/musicvg/playlistlv
Rounding size to boundary between physical extents: 52.00 MiB.
Size of logical volume musicvg/playlistlv changed from 300.00 MiB (75 extents) to 352.00 MiB (88 extents).
Logical volume musicvg/playlistlv successfully resized.
[root@localhost ~]# resize2fs /dev/musicvg/playlistlv
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/musicvg/playlistlv is mounted on /mnt/playlist; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 3
The filesystem on /dev/musicvg/playlistlv is now 360448 (1k) blocks long.

[root@localhost ~]# df -h
Filesystem              Size  Used Avail Use% Mounted on
devtmpfs                 3.8G   0  3.8G   0% /dev
tmpfs                   3.8G   0  3.8G   0% /dev/shm
tmpfs                    1.5G  9.7M  1.5G   1% /run
/dev/mapper/cs-root       36G   5.5G   30G  16% /
/dev/nvme0n1p1           960M  468M  493M  49% /boot
tmpfs                    765M   92K  765M   1% /run/user/0
/dev/sr0                  11G   11G   0 100% /run/media/root/CentOS-Stream-9-BaseOS-x86_64
/dev/mapper/musicvg-playlistlv 321M   14K   300M   1% /mnt/playlist
[root@localhost ~]#
```